

Do **not** write solutions on this page.

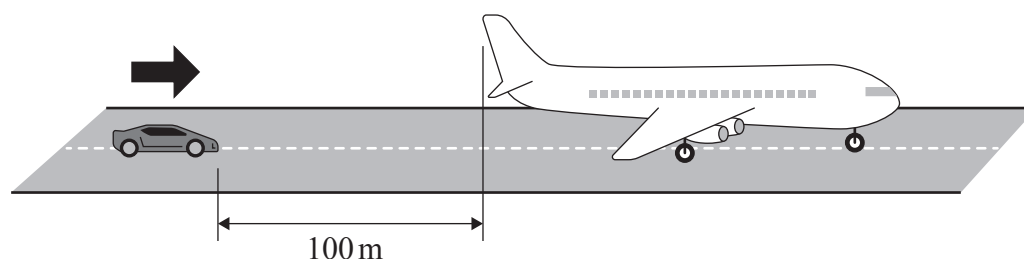
Section B

Answer **all** questions in the answer booklet provided. Please start each question on a new page.

10. [Maximum mark: 17]

An airplane lands on a runway 100 metres in front of a stationary car. At the instant the airplane lands, the car begins to travel in the same direction towards the airplane.

diagram not to scale



Let t represent the number of seconds after the airplane lands. For $t \geq 0$, the velocities of the airplane and the car in m s^{-1} , can be modelled by the following equations:

$$v_{\text{air}} = 60e^{-0.1t}$$

$$v_{\text{car}} = 5t$$

(a) When the airplane lands, write down the speed of

(i) the airplane;

(ii) the car.

[2]

(b) Find

(i) the value of t when the airplane and the car have the same speed;

(ii) the speed at this time.

[3]

Let $d(t)$ represent the distance, in metres, between the car and the back of the airplane after t seconds.

(c) If $d(0) = 100$, find $d(t)$.

[7]

(d) Hence, find how long it takes for the car to reach the back of the airplane.

[2]

(e) Find the distance travelled by the car when it reaches the back of the airplane.

[3]

